Non-Linear Dynamics

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Certificate from the Supervisor

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1 Introduction

Nonlinear dynamics is a rich and fascinating area of mathematics and applied science that explores systems governed by nonlinear relationships. Unlike linear systems, where changes in inputs produce proportional changes in outputs, nonlinear systems can exhibit a wide variety of complex behaviors, including multistability, limit cycles, bifurcations, and chaos. These behaviors are often observed in real-world systems ranging from ecological models and fluid dynamics to electrical circuits and economic processes.

This project focuses on the analysis of chaotic behavior in nonlinear dynamical systems. Chaos, in this context, refers to deterministic yet unpredictable behavior that arises due to high sensitivity to initial conditions — a hallmark of nonlinear systems. Even though such systems are governed by well-defined equations, their long-term behavior can appear random and disordered.

In this study, we investigate a specific nonlinear model — a [brief description of the system, e.g., three-species population model with nonlinear functional responses] — and explore how chaos emerges as parameters are varied. We analyze the system's behavior through bifurcation analysis, time series, and phase portraits, aiming to identify the conditions under which chaotic dynamics appear.

By understanding the routes to chaos in such systems, we gain deeper insights into the complex behavior that nonlinear models can produce, which is crucial for both theoretical development and real-world applications where prediction and control are essential.

2 Nonlinear Dynamics

Nonlinear dynamics focuses on systems where the output is not directly proportional to the input, meaning a small change in the input can lead to a disproportionately large change in the output, or even a qualitative large change in behaviour.

2.1 Definition

The study of systems whose behaviour is not governed by simple, linear equations, often resulting in complex, unpredictable and seemingly chaotic patterns. Such a system is called a nonlinear dynamical system.

2.2 Examples

1. Weather
2. Population Dynamics
3. Heartbeat
4. Stock Market
5. Anharmonic Oscillator

2.3 Types of Dynamical Systems

2.3.1 Continuous Dynamical System

A Continuous Dynamical system is a system whose state evolves over state space continuously according to a fixed rule. It is represented by differential equations.

Example

$$\dot{x} = ax - xy,$$
$$\dot{y} = xy - by.$$

Another example:

$$\ddot{x} + \alpha\dot{x} + \omega_0^2 x + \beta x^3 = f\cos\omega t.$$

2.3.2 Discrete Dynamical System

A Discrete Dynamical system is a system whose state evolves over state space discontinuously according to a fixed rule. It is represented by an iterative map.

Example

1. $X_{n+1} = aX_n(1 - X_n), \quad a > 0.$
2. $X_{n+1} = 1 - aX_n^2 + Y_n,$
   $Y_n = bX_n, \quad a, b > 0.$

3 Stability Analysis

3.1 Continuous case:

These are systems that evolve over time, and their behavior is described by equations that relate the current state to the next state, which is called a *dynamical system*
The qualitative change of any dynamical system with respect to the change of one or more than one system parameter is called *bifurcation.*
The particular Point of Parameter/Parameters at which bifurcation occurs is called *critical/bifurcation point* of the parameter and the parameter/parameters with respect to which bifurcation occurs is called *bifurcation parameter*

When a system doesn't change its dynamical status that means when there is no rate of Change of System that means the input and output of the system is same.Then the system is called *Equilibrium* and the state is called *Equilibrium State.*

Phase Space
Diagram of different phases of a dynamical system is called a *phase space diagram.*
We plot the diagram with respect to time (or the independent variable of the dynamical system). Then it is called *time series* diagram.In that kind of diagram time or the independent variable of the dynamical system is executed explicitly.
When we plot the diagram with respect to the Dependent variable of the dynamical system, where the time(or the independent variable of the dynamical system) is not executed explicitly.Then it is called *phase plane(for 2-D)/Phase portrait.*
Trajectories are called *Phase Trajectories.*

3.1.1 General Criteria For Stability(2-D Case)

General form of 2D or 2-Dim Autonomous Dynamical System is written in the Form

$$\frac{dx}{dt} = P(x, y) \tag{1}$$

$$\frac{dy}{dt} = Q(x, y) \tag{2}$$

where P and Q are some well defined function of x and y and constant parameters
Suppose E*(x*,y*) is the equilibrium point of the system
Let X= x-x*
Y= y-y*
So the System can be written as

$$\frac{d(X + x*)}{dt} = P(X + x*, Y + y*) \tag{3}$$

$$\frac{dX}{dt} + \frac{dx*}{dt} = P(X + x*, Y + y*) \tag{4}$$

But x* is the Equilibrium point hence constant so

$$\frac{dx^*}{dt} = 0 \tag{5}$$

Hence we get

$$\frac{dX}{dt} = P(X + x^*, Y + y^*) \tag{6}$$

Similarly

$$\frac{dY}{dt} = Q(X + x^*, Y + y^*) \tag{7}$$

So the System will be

$$\frac{dX}{dt} = P(X + x^*, Y + y^*) \tag{8}$$

$$\frac{dY}{dt} = Q(X + x^*, Y + y^*) \tag{9}$$

Now we will apply Taylor Series Expansion with respect to x* and y* in equation (8) and equation(9)

$$\begin{aligned}
\frac{dX}{dt} &= P(X + x^*, Y + y^*) \\
&= P(x^*, y^*) + X\left(\left.\frac{\partial P}{\partial X}\right|_{(x^*,y^*)}\right) + Y\left(\left.\frac{\partial P}{\partial Y}\right|_{(x^*,y^*)}\right) + H(0) \\
&= X\left(\left.\frac{\partial P}{\partial X}\right|_{(x^*,y^*)}\right) + Y\left(\left.\frac{\partial P}{\partial Y}\right|_{(x^*,y^*)}\right) \\
&= a_{11}X + a_{12}Y
\end{aligned}$$

(Here (x*,y*) is the equilibrium point so P(x*,y*)=0)
Similarly

$$\begin{aligned}
\frac{dX}{dt} &= Q(X + x^*, Y + y^*) \\
&= Q(x^*, y^*) + X\left(\left.\frac{\partial Q}{\partial X}\right|_{(x^*,y^*)}\right) + Y\left(\left.\frac{\partial Q}{\partial Y}\right|_{(x^*,y^*)}\right) + H(0) \\
&= X\left(\left.\frac{\partial Q}{\partial X}\right|_{(x^*,y^*)}\right) + Y\left(\left.\frac{\partial Q}{\partial Y}\right|_{(x^*,y^*)}\right) \\
&= a_{21}X + a_{22}Y
\end{aligned}$$

(Here (x*,y*) is the equilibrium point so Q(x*,y*)=0)

So the system will be

$$\begin{aligned}
\frac{dx}{dt} &= a_{11}X + a_{12}Y \\
\frac{dy}{dt} &= a_{21}X + a_{22}Y
\end{aligned}$$

We can write the above equation as Matrix Form

$$\frac{d}{dt}\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

so the above matrix can be Written as $\dot{X}$=JX
Where

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

and the Jacobian Matrix

$$J = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Then we determine the eigenvalues from the characteristic equation of the above Jacobian matrix . Let the eigenvalues are λ_1 and λ_2 . Substituting the equilibrium points in the eigenvalues we get Stability nature of equilibrium point, which given below in table:

Stability nature of equilibrium point	**Conditions on the eigenvalues**
Stable	Real parts of both the eigenvalues are negative
Unstable	Real part of even one of the two eigenvalues is positive
Neutral	Real parts of both the eigenvalues are zero

3.2 Discrete case

If x is a fixed point and satisfies $f(x^*) = x^*$, we can write:

$$x_{n+1} = f(x_n) = f(x^* + \eta_n) \tag{10}$$

To determine the stability of x, we consider a nearby η_n such that $x_n = x^* + \eta_n$. Expanding in Taylor series,

$$x_{n+1} = f(x^* + \eta_n) = f(x^*) + f'(x^*)\eta_n + O(\eta_n^2) \tag{11}$$

Since $f(x^*) = x^*$, the above equation simplifies to:

$$\eta_{n+1} = f'(x^*)\eta_n \tag{12}$$

Neglecting $O(\eta_n^2)$, we get:

$$\eta_n = f'(x^*)^n \eta_0 \tag{13}$$

So, after n iterations:

$$\eta_n = f'(x^*)^n \eta_0 \tag{14}$$

If $|f'(x^*)| < 1$, then $\eta_n \to 0$, meaning the fixed point is stable. If $|f'(x^*)| > 1$, then η_n grows, meaning the fixed point is unstable

4 Attractors

4.1 Definition

An attractor is a region in the phase space to which nearby trajectories tend to converge, representing the long-term behavior of the system.

4.2 Types of Attractors

1. **Point Attractor**: In nonlinear dynamics, a point attractor is a stable equilibrium state toward which a system evolves over time. Regardless of initial conditions (within a certain basin of attraction), the system's trajectory eventually settles at this fixed point.

2. **Limit Cycle Attractor**:A limit cycle attractor is a closed, isolated trajectory in the phase space of a nonlinear dynamical system, toward which nearby trajectories asymptotically converge over time. Unlike fixed-point attractors, which lead to equilibrium states, a limit cycle represents periodic behavior that persists under small perturbations. A limit cycle attractor satisfies:

$$X(t) = X(t+T), \quad 0 < T < \infty.$$

   A stable limit cycle is a period-T attractor or a periodic attractor

3. **Quasiperiodic Attractor:**More specifically, a quasiperiodic orbit is one which can be written as

$$X(t) = \phi(\omega_1 t, \omega_2 t, \ldots, \omega_n t), \tag{15}$$

   where ϕ is periodic, of period 2π, in each of the $n(n > 1)$ arguments. Further, the frequencies $(\omega_1, \omega_2, \ldots, \omega_n)$ must have the following properties:

   (a) They are linearly independent, that is, there does not exist a nonzero set of integers $(l_1, l_2, \ldots, l_n)$ such that

$$l_1\omega_1 + l_2\omega_2 + \cdots + l_n\omega_n = 0. \tag{16}$$

   (b) For each i, $\omega_i \neq |l_1\omega_1 + l_2\omega_2 + \cdots + l_n\omega_n|$ for some integers $(l_1, l_2, \ldots, l_n)$, that is, there is no rational relation between the ω's. (For example, for $n = 2$, this implies that the ratio ω_1/ω_2 is irrational.)

   In other words, a *quasiperiodic orbit* is the sum of periodic orbits each of whose frequency is one of the various sums and differences of the finite set of frequencies $(\omega_1, \omega_2, \ldots, \omega_n)$. Such a quasiperiodic motion takes place on a n-dimensional torus $\mathbb{T}^n$. Suppose that this torus is an attracting set, that is nearby trajectories get attracted to it, then we say that $\mathbb{T}^n$ is a *quasiperiodic attractor.*

   As a simple example for the realization of quasiperiodic motion let us assume that the approximate solution of a dynamical system

$$\ddot{x} = F(x, \dot{x}) + f\cos\omega_1 t, \tag{17}$$

where F is nonlinear, is given by

$$x(t) = b_1 \cos \omega_1 t + b_2 \cos \omega_2 t. \tag{18}$$

The motion is periodic if the frequency ratio ω_1/ω_2 is a rational number. Consequently, the solution curve in the $(x, y = \dot{x})$ phase space is a simple closed orbit. If ω_1/ω_2 is irrational then the motion is not periodic. Figure 3.10a shows the phase portrait for $b_1 = 1$, $b_2 = 1$, $\omega_1 = 1$, $\omega_2 = \sqrt{2}$. The solution is clearly not a closed orbit and is *quasiperiodic* or almost periodic in the sense that trajectory returns arbitrarily close to its starting point infinitely often.

4. **Strange Attractor or Chaotic Attractor**:A strange attractor is a type of attractor in nonlinear dynamical systems that has a fractal structure and exhibits chaotic behavior. Unlike simple attractors (fixed points or limit cycles), strange attractors are non-periodic and display sensitive dependence on initial conditions, meaning small differences in starting points lead to vastly different outcomes.

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5 Definition of Chaos

5.1 What Chaos Actually Is?

Weather is one of the best real-life examples of chaos.Short-term weather forecasts (1-2 days) are usually accurate, but long-term weather forecasts (weeks or months) become unreliable because tiny, unmeasurable changes grow and influence the entire system. A slight difference in temperature, humidity, or wind speed can drastically change future weather patterns. Two locations with nearly identical weather today can have completely different conditions a week later. Lorenz discovered this effect while working on weather prediction models. He found that rounding off decimal points in his calculations led to wildly different long-term forecasts.This showed that deterministic systems (systems following fixed rules) could still produce unpredictable results. Even with modern supercomputers and advanced models, weather predictions are still imperfect. This is because:We cannot measure every tiny detail in the atmosphere.Small, unobserved changes in one place can grow and influence global weather. This unpredictable change in the result with respect to the initial condition is called chaos.

5.2 Definition

5.2.1 Formal Definition of Chaos

Chaos is a property of certain dynamical systems that exhibit extreme sensitivity to initial conditions, leading to seemingly random and unpredictable long-term behavior despite being governed by deterministic rules.

5.2.2 Mathematical Definition

A dynamical system is said to be **chaotic** if it satisfies the following three conditions:

1. **Sensitive Dependence on Initial Conditions (Butterfly Effect)**

- Small differences in the initial state of the system lead to exponentially large differences in future states.
- Mathematically, this is often expressed using the **Lyapunov exponent** $\lambda > 0$, which quantifies how quickly nearby trajectories diverge:

$$d(t) \approx d(0)e^{\lambda t}$$

where $d(t)$ is the distance between two initially close trajectories.

2. **Topological Transitivity (Mixing Behavior)**
   - The system evolves in such a way that any given region of the phase space will eventually overlap with any other given region.
   - This ensures that the system explores all possible behaviors over time.
3. **Dense Periodic Orbits**
   - Within the chaotic system, there exist infinitely many periodic orbits that are arbitrarily close to any given point in the phase space.
   - This means that the system has both structured (periodic) and unpredictable behavior.

6 Routes to Chaos

6.1 Period Doubling Route to Chaos

A system that originally oscillates with a single period begins to show oscillations with twice that period, then four times, eight times, and so on — eventually leading to chaos.

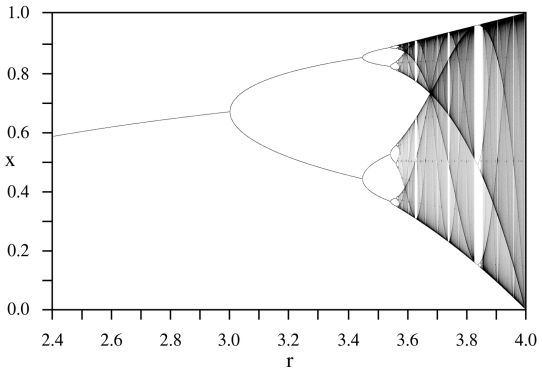


Figure 1: Period Doubling

6.2 Quasiperiodic Route to Chaos

Quasiperiodic route to chaos follows the Ruelle-Takens-Newhouse scenario, where the system undergoes multiple Hopf bifurcations leading to a quasiperiodic orbit that eventually transitions to chaos.

System starts oscillating with two or more incommensurate (non matching) frequencies.These frequencies interfere and create a complex pattern that eventually leads to chaos.Common in systems with multiple oscillatory components

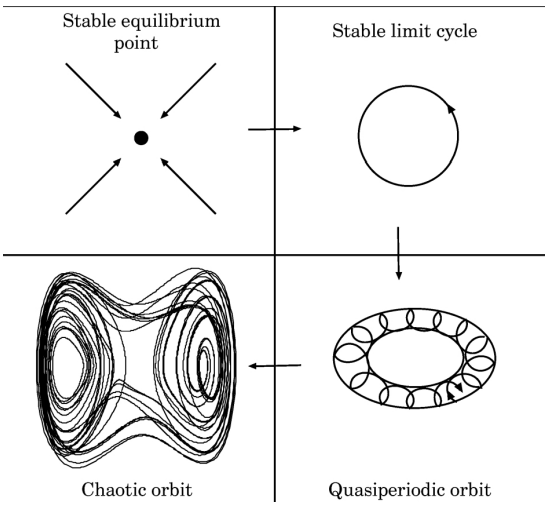


Figure 2: Quasiperiodic Route to Chaos

6.3 Intermittency Route to Chaos

The Pomeau-Manneville scenario describes intermittency route to chaos where the motion alternates between regular laminar phases and irregular bursts, leading to fully chaotic behavior as a control parameter increases.

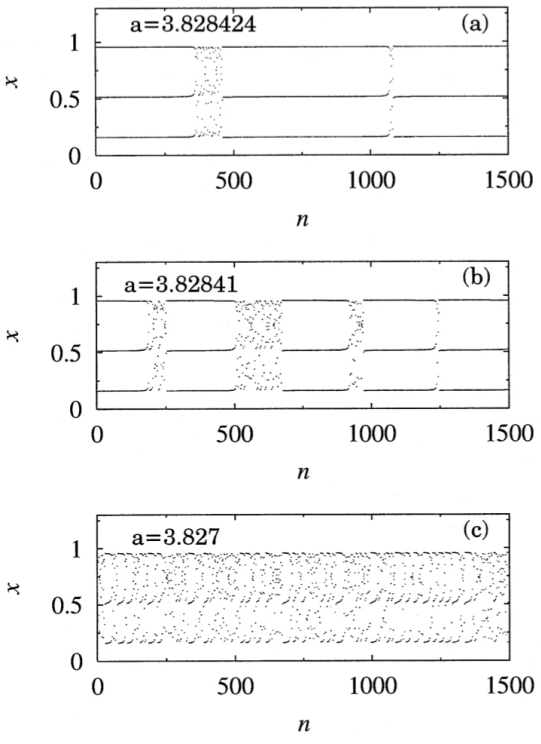


Figure 3: Intermittency Route to Chaos

7 Chaos in Continuous Dynamical System

7.1 Lorenz Equation

We all know what is Bifurcation, that is the qualitative change of the dynamics of the system with respect to the change of some parameter (Like creation or destruction of fixed points).

There are various kinds of bifurcations like saddle-node, transcritical, pitchfork, Hopf, and so on. And we all know how the system behaves in these kinds of bifurcations with changing parameters.

But in 1963, Ed Lorenz derived a set of equations for a 3-D model of convection roll in the atmosphere, which are:

$$\dot{x} = \sigma(y - x) \tag{3}$$
$$\dot{y} = rx - y - xz \tag{4}$$
$$\dot{z} = xy - bz \tag{5}$$

In the above equations, there are three parameters: σ, r, b.

Lorenz observed that these equations show more than bifurcations. Over a large range of parameters, they exhibit very strange dynamics. Their solution is oscillating aperiodically but still bounded. Such a solution set is now called a **Strange Attractor**.

8 Solution of Lorenz Equations

As we can see clearly from the above example that the Lorenz equations lead to chaos, we will solve it to see how parameter changes lead the system to chaos.

Lorenz equations are given by:

$$\begin{cases} \dot{x} = \sigma(y - x), \\ \dot{y} = rx - y - xz, \\ \dot{z} = xy - bz. \end{cases} \tag{19}$$

To find the fixed points of the system, we have $\dot{x} = 0, \dot{y} = 0, \dot{z} = 0$. So we get from Eq. (19), (20), (21):

$$x = y \tag{22}$$
$$rx = y + xz \tag{23}$$
$$bz = xy \tag{24}$$

From these, we get the fixed points:

$$E_0 = (0, 0, 0) \tag{25}$$
$$E_1 = (\sqrt{b(r-1)}, \sqrt{b(r-1)}, r - 1) \tag{26}$$
$$E_2 = (-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r - 1) \tag{27}$$

We will take $b = 8/3$ and $\sigma = 10$ to get the chaos for variations in the parameter r.

The general form of the Jacobian matrix is given by:

$$J = \begin{bmatrix} -\sigma & \sigma & 0 \\ r - z & -1 & -x \\ y & x & -b \end{bmatrix} \tag{28}$$

Case 1: For $r < 1$, E_1 and E_2 will not exist, as seen from Eq. (26) and (27). So we have only one fixed point, E_0.

Now, the Jacobian matrix of the Lorenz Eq. at E_0 is given by:

$$J_{E_0} = \begin{bmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{bmatrix} \tag{29}$$

Now, with $\sigma = 10$ and $b = 8/3$, the Jacobian at $(0, 0, 0)$ is:

$$J_{E_0} = \begin{bmatrix} -10 & 10 & 0 \\ r & -1 & 0 \\ 0 & 0 & -8/3 \end{bmatrix} \tag{30}$$

Now, we have to evaluate the eigenvalues of J:

$$(10 - \lambda)(1 + \lambda)(\lambda + 8/3) + 10(8/3 + \lambda)r = 0 \tag{31}$$

Solving this, we obtain:

$$\lambda_1 = -8/3, \quad \lambda_2 = \frac{-11 + \sqrt{81 + 40r}}{2}, \quad \lambda_3 = \frac{-11 - \sqrt{81 + 40r}}{2} \tag{32}$$

From Eq. (32), for $r < 1$, all eigenvalues are negative, so $(0, 0, 0)$ is a stable node.

Case 2: At $r = 1$, the eigenvalues are:

$$\lambda_1 = -b = -8/3, \quad \lambda_2 = -1, \quad \lambda_3 = 0 \tag{33}$$

This indicates that two eigen-directions are stable, but one is trying to be unstable. Though $(0, 0, 0)$ becomes unstable for $r > 1$, at $r = 1$, eigenvalues are still real. Thus, E_0 becomes a saddle point.

Case 3: For $r > 1$, E_1 and E_2 both exist, and we check the stability of $(0, 0, 0)$. The Jacobian at E_1 is:

$$J_{E_1} = \begin{bmatrix} -10 & 10 & 0 \\ \sqrt{b(r-1)} & -1 & -\sqrt{b(r-1)} \\ \sqrt{b(r-1)} & \sqrt{b(r-1)} & -8/3 \end{bmatrix} \tag{34}$$

For $r > 1$, checking the nature of the fixed point for $r = 2.8$:

$$J_{E_1} = \begin{bmatrix} -10 & 10 & 0 \\ \sqrt{8/3(2.8-1)} & -1 & -\sqrt{8/3(2.8-1)} \\ \sqrt{8/3(2.8-1)} & \sqrt{8/3(2.8-1)} & -8/3 \end{bmatrix} \tag{35}$$

Solving, we obtain:

$$\lambda = \frac{-11 \pm \sqrt{81 + 40r}}{2} \tag{36}$$

From Eq. (36), we get one real eigenvalue and two complex conjugate eigenvalues with a real part.

Eigenvector Calculation: Let:

$$\dot{x} = \sigma x - xy \tag{39}$$

$$\dot{y} = rx + xy \tag{40}$$

The characteristic equation is given by:

$$|J - \lambda I| = 0 \tag{41}$$

So eigenvalues are:

$$\lambda_1 = \sigma + j\omega, \quad \lambda_2 = \sigma - j\omega \tag{20}$$

Eigenvector corresponding to $\lambda_1 = \sigma + j\omega$:

$$\begin{bmatrix} \sigma - j\omega & 0 \\ 0 & \sigma - j\omega \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \tag{43}$$

Solving:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ j \end{bmatrix} \tag{44}$$

Similarly, for $\lambda_2 = \sigma - j\omega$, the eigenvector is:

$$\begin{bmatrix} 1 \\ -j \end{bmatrix} \tag{21}$$

From here we can say that for complex conjugate eigenvalues, eigenvectors are also complex conjugate. So the solution of Eq. (39) and (40) can be written as:

$$x(t) = e^{\sigma t} e^{j\omega t} \begin{pmatrix} j \end{pmatrix} \tag{45}$$

$$x(t) = e^{\sigma t} \begin{pmatrix} \cos(\omega t) - j\sin(\omega t) \\ \sin(\omega t) + j\cos(\omega t) \end{pmatrix} \tag{46}$$

$$x(t) = e^{\sigma t} \begin{pmatrix} -\sin(\omega t) \\ \cos(\omega t) \end{pmatrix} + j e^{\sigma t} \begin{pmatrix} \cos(\omega t) \\ \sin(\omega t) \end{pmatrix} \tag{47}$$

$$x(t) = C_1 e^{\sigma t} \begin{pmatrix} -\sin(\omega t) \\ \cos(\omega t) \end{pmatrix} + C_2 e^{\sigma t} \begin{pmatrix} \cos(\omega t) \\ \sin(\omega t) \end{pmatrix} \tag{48}$$

From the above Eq. (48), it is clear that $x(t)$ is a linear combination of two independent basis vectors, so it gives a plane. Thus, we conclude that complex conjugate eigenvalues describe a plane.

Using the above result, we see from Eq. (38) that we get one real and two complex conjugate eigenvalues. This means they define a plane containing E_1. Additionally, due to the one negative real eigenvalue, there will exist a stable eigen-direction (passing through E_1) along which the solution will decay exponentially towards E_1. Similarly, for the other fixed point E_2, the same result holds.

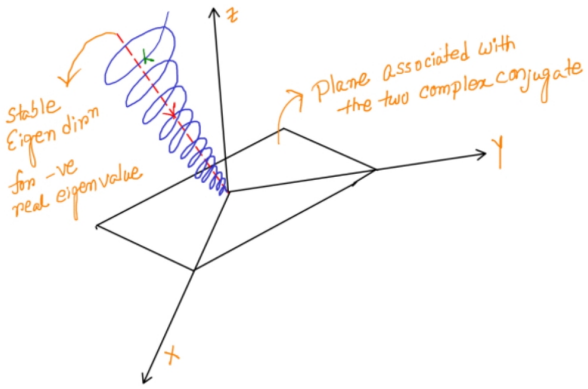


Figure 4: Phase space trajectory for two complex conjugate and one real eigenvalue

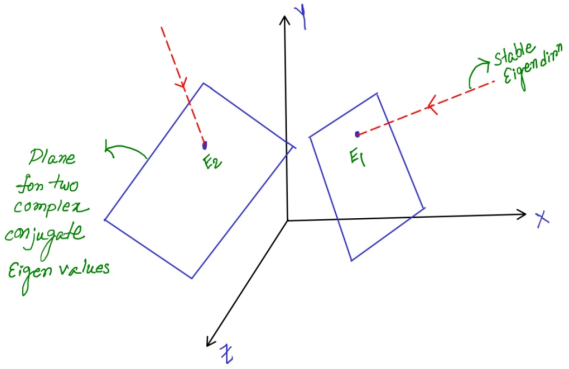


Figure 5: Phase planes for Lorenz Equation

9 Plot of Lorenz Equation at Phase Space for Different r Values

1. **For $r < 1$, we see only one point E_0 exists, which is a stable node. Any initial condition will decay to E_0 over time.**

2. **For $r = 1$, a transition occurs where the stable E_0 becomes unstable. At this point, two stable branches also emerge, forming a pitchfork bifurcation.**

3. **For $r > 1$, E_0 becomes unstable. Any initial condition will move towards one of the two stable spiral points, E_1 or E_2. The structure of these fixed points consists of a stable point encircled by an unstable limit cycle. This pattern continues until $r = r_H$, at which E_1 and E_2 lose stability in a Hopf bifurcation.**

4. **For $r > r_H$, due to subcritical Hopf bifurcation, the stable fixed points E_1 and E_2 absorb the saddle cycle and turn into saddle points. Beyond $r > r_H$, there are no more attractors in the neighborhood. In principle, for $r > r_H$, trajectories should fly away to a distant attractor.**

   However, Lorenz argued that for $r > r_H$, the existing limit cycle must become unstable, leading to chaotic behavior.

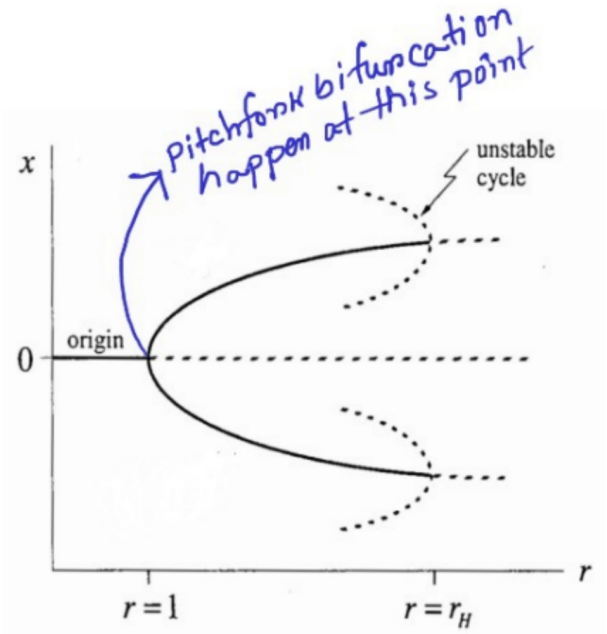


Figure 6: Bifurcation Diagram

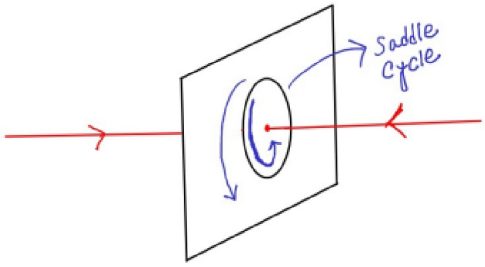


Figure 7: Saddle Cycle at $r < r_H$

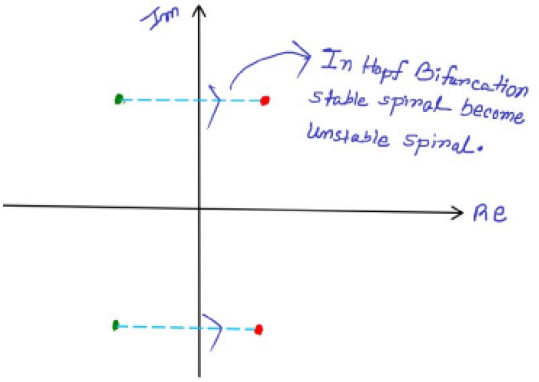


Figure 8: Hopf Bifurcation in Linearized Region

5. **Lorenz took parameter values $\sigma = 10$, $b = 8/3$, and $r = 28$. At this parameter set, he observed chaos just past the Hopf bifurcation value $r_H \approx 24.74$. The chaotic pattern is shown in the figure below.**

10 Chaos in Discrete Dynamical System

10.1 Logistic Map

I am taking a simple example of a discrete dynamical system (Logistic Map) to describe chaos. The equation has the following form:

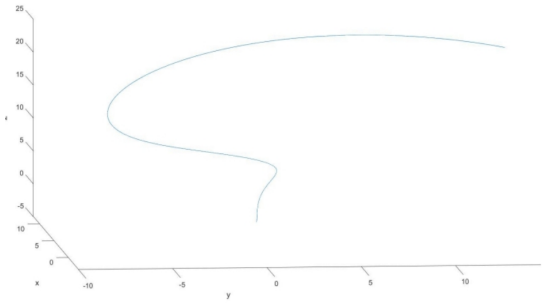


Figure 9: For r = 0.2

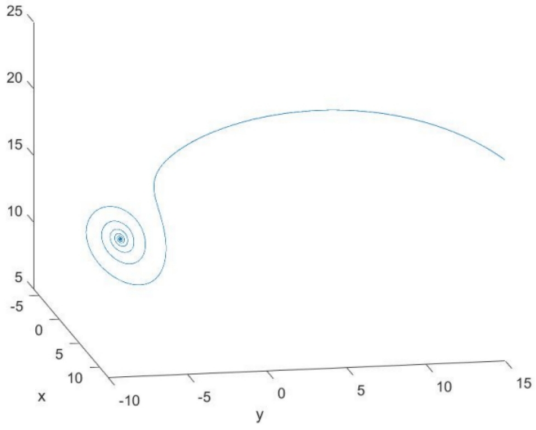


Figure 10: For r = 12

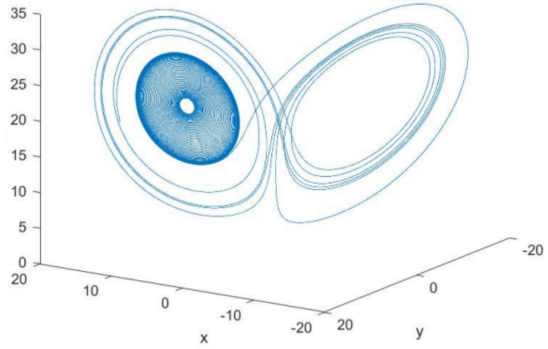


Figure 11: For r = 22

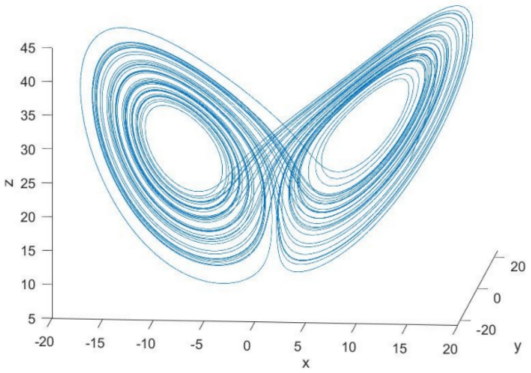


Figure 12: For r = 28

$$x_{n+1} = \mu x_n(1 - x_n) \tag{22}$$

For a discrete dynamical system, a fixed point corresponding to a period-1 orbit can be evaluated by the condition:

$$x_n = x_{n+1} \tag{23}$$

So, for the Logistic Map, fixed points are:

$$x_0^* = 0, \quad x_1^* = 1 - \frac{1}{\mu} \tag{24}$$

11 Period Doubling Cascade

$$\frac{dx_{n+1}}{dx_n} = \mu; \quad \textbf{for } x_n^* = 0 \tag{57}$$

$$\frac{dx_{n+1}}{dx_n} = 2 - \mu; \quad \textbf{for } x_n^* = \frac{\mu - 1}{\mu} \tag{58}$$

From Eq. (57) we can see if $\mu < 1$ then $x_n^* = 0$ is a stable point.

From Eq. (57) we can see if $\mu > 1$ then $x_n^* = 0$ is an unstable point.

Similarly, from Eq. (58) we can see the range of stability $1 < \mu < 3$. So at $\mu > 1$, the origin loses stability and the other fixed point becomes stable. You can see the corresponding cobweb diagram from Fig. (19) and (20).

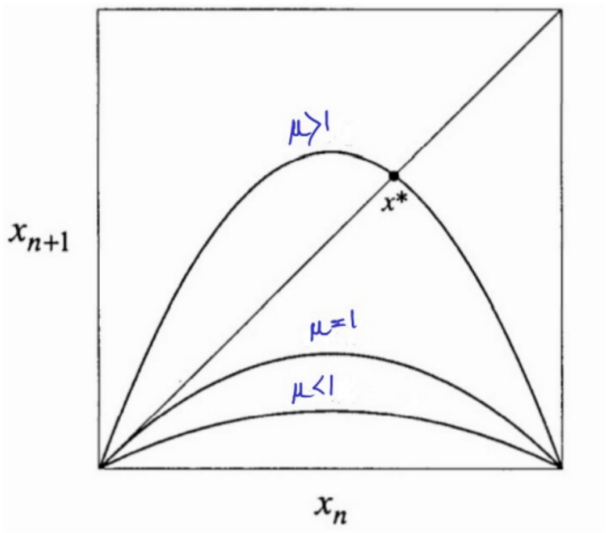


Figure 13: Change of Fixed Point with Parameter

So from here we can conclude Period-1 orbit is stable up to $\mu = 3$. After that, the Period-1 orbit becomes unstable, and the Period-2 orbit becomes stable. Period-1 orbit starts to lose stability when the slope of the x_{n+1} vs. x_n graph becomes -1.

And the condition for the period-2 orbit is:

$$x_{n+2} = x_n$$

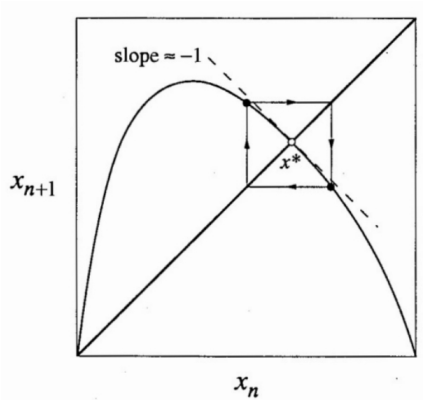


Figure 14: Loss of Stability at -1 Slope

For a period-2 orbit, we set:

$$x_{n+1} = \mu x_n(1 - x_n) \tag{25}$$
$$x_{n+2} = \mu x_{n+1}(1 - x_{n+1}) \tag{26}$$

Applying this condition:

$$x_{n+2} = \mu x_n(1 - x_n) \cdot (1 - \mu x_n(1 - x_n)) \tag{27}$$

Since the fixed points of the period-1 orbit are also fixed points of the period-2 orbit, two roots of this equation are already known. Thus, we solve for the additional period-2 orbit points:

$$x_n(1 - \mu + \mu x_n)(1 + \mu - \mu^2 x_n + \mu^2 x_n^2) = 0 \tag{28}$$

Solving this,

$$x_3^* = \frac{(1+\mu) - \sqrt{(1+\mu)(\mu-3)}}{2\mu}, \quad x_3^* = \frac{(1+\mu) + \sqrt{(1+\mu)(\mu-3)}}{2\mu} \tag{29}$$

For $\mu > 3$, real solutions exist, indicating the onset of period-2 orbits. We define:

$$x_{n+1} = f(x_n) \tag{30}$$

$$x_{n+2} = f(f(x_n)) = f^2(x_n) \quad \textbf{and so on} \tag{31}$$

Now, the question is whether the period-2 orbit is stable or not. The stability of the period-2 orbit is determined by the derivative:

$$\left| \frac{dx_{n+2}}{dx_n} \right| \tag{32}$$

which is given by:

$$\frac{dx_{n+2}}{dx_n} = f'(f(x_n))f'(x_n) \tag{33}$$

So, this is the composition of slopes at the two points which are the fixed points of the period-2 orbit. If:

$$\left| \frac{dx_{n+2}}{dx_n} \right| < 1 \tag{34}$$

then the period-2 orbit is stable. From Eq. (60), we get:

$$\begin{aligned} \left| \frac{dx_{n+2}}{dx_n} \right| &= |\mu(1 - 2x_3^*)\mu(1 - 2x_4^*)| \\ &= |\mu^2[1 - 2(x_3^* + x_4^*) + 4x_3^* x_4^*]| \\ &= \left| \mu^2 \left[1 - 2\frac{(\mu+1)}{\mu} + 4\frac{(\mu+1)}{\mu^2} \right] \right| \\ &= \left| 4 + 2\mu - \mu^2 \right| \end{aligned}$$

So, the period-2 orbit is stable for:

$$|4 + 2\mu - \mu^2| < 1 \tag{35}$$

The range of stability is:

$$3 < \mu < 1 + \sqrt{6} \tag{36}$$

This process goes on indefinitely, leading to a period-doubling cascade that ultimately results in chaos.

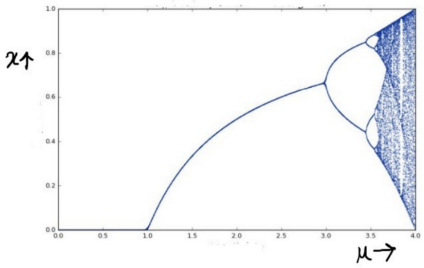


Figure 15: Period Doubling Chaos

11.1 Cobweb Diagram

Here, I show how the cobweb diagram changes with the parameter μ and how it takes shape at different periodic orbits. In Fig. (19) and (20), you can see that up to $\mu = 3$, the period-1 orbit is stable and how stability changes from one fixed point to the other. If you increase μ further, the period-1 orbit loses stability, leading to a period-2 orbit (Fig. 18), then a period-4 orbit (Fig. 21), and ultimately chaos (Fig. 22).

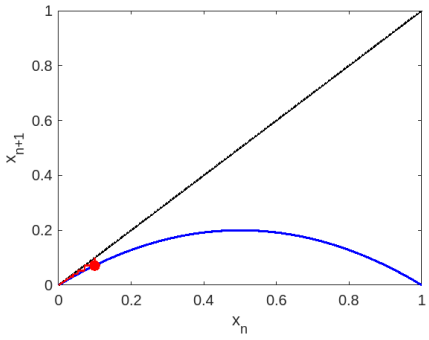


Figure 16: For $\mu < 1$

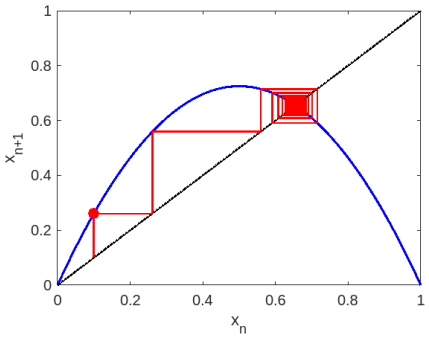


Figure 17: For $1 < \mu < 3$

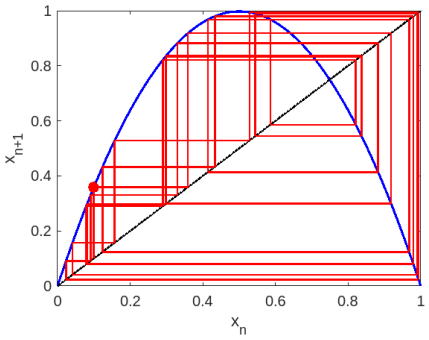


Figure 18: For Chaos

12 Applications of Chaos Theory

Chaos, despite its unpredictable nature, has many practical applications across different fields. Here are a few key examples:

1. **Cryptography**
    - **Chaos theory is used in encryption algorithms to create secure communication channels.**
    - **Chaotic maps generate pseudo-random sequences for secure key generation.**
2. **Weather Prediction**
    - **Meteorologists use chaos theory to model atmospheric systems, helping improve weather forecasts.**
    - **Small changes in initial conditions can significantly impact predictions, guiding probabilistic models.**
3. **Stock Market & Economics**
    - **Chaos theory helps analyze financial markets, which exhibit nonlinear and unpredictable behaviors.**
    - **Traders use it to model economic cycles and price fluctuations.**

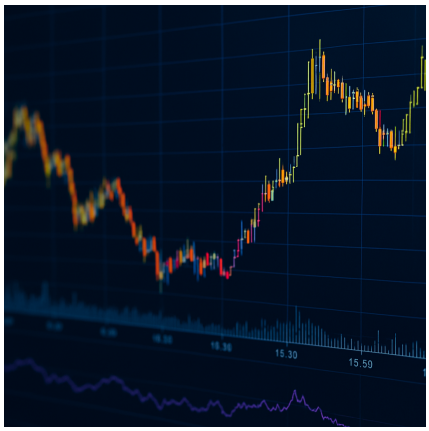


Figure 19: Chaos in stock price

    -
4. **Artificial Intelligence & Machine Learning**
    - **Neural networks use chaos-inspired approaches for training and optimization.**
    - **Genetic algorithms incorporate chaotic elements to improve search efficiency.**
5. **Traffic Flow & Urban Planning**

- **Chaos theory helps model traffic congestion and optimize urban infrastructure.**
- **Small interventions (like adjusting traffic lights) can significantly impact overall flow.**

6. **Medicine & Biology**
    - **In cardiology, chaotic models help understand irregular heartbeats (arrhythmias).**

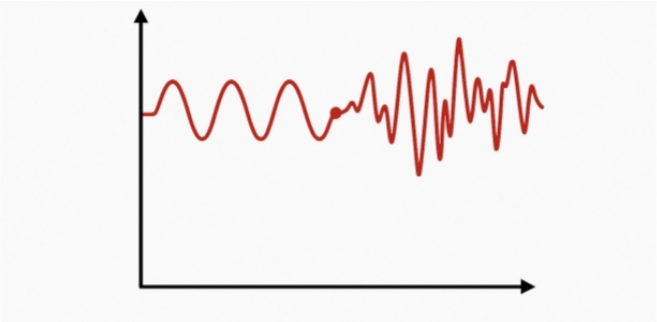


Figure 20: Chaotic behavior before heart attack

- **Epidemiology uses chaos to predict disease spread under different conditions.**

13 Three species non-delayed model system

We will analyze a model that describes a tritrophic food chain composed of a prey, whose population density is denoted by X, a predator whose population density is denoted by Y and a superpredator (or top predator), whose population density is denoted by Z. Let us introduce a tritrophic food chain model within the framework of the following set of nonlinear ordinary differential equations:

$$\frac{dX}{dt} = X\left(a_1\left(1-\frac{X}{K}\right) - \frac{wYX^2}{\frac{X^2}{j}+X+a}\right), \tag{37}$$

$$\frac{dY}{dt} = Y\left(-a_2 + \frac{w_1X}{\frac{X^2}{j}+X+a} - \frac{w_2Z}{Y+D_2+bZ}\right), \tag{38}$$

$$\frac{dZ}{dt} = Z\left(-c + \frac{w_3Y}{Y+D_2+bZ}\right). \tag{39}$$

13.1 Equilibrium Points of the Model System

- $E_0(0,0,0)$,
- $E_1(K,0,0)$,
- $E_2(x_2,y_2,0)$: **where** x_2 **and** y_2 **are defined as**

$$x_2 = \frac{\left(\frac{\omega_1}{a_2}-1\right) \pm \sqrt{\left(\frac{\omega_1}{a_2}-1\right)^2 - 4aj}}{2}, \quad y_2 = \frac{a_1}{\omega}\left(1-\frac{x_2}{K}\right)\left(\frac{\omega_1 x_2}{a_2}\right) = Nx_2\left(1-\frac{x_2}{K}\right).$$

- $E_3(x_3,y_3,z_3)$: **where** x_3, y_3 **and** z_3 **are defined as**

$$y_3 = \frac{a_1}{\omega}\left(1-\frac{x_3}{K}\right)\left(\frac{x_3^2}{j}+x_3+a\right), \quad z_3 = \frac{y_3(\omega_3-c)-cD_2}{bc}$$

such that $K > x_3$, $\omega_3 > c$, **and** $y_4 > \frac{cD_2}{\omega_3-c}$.

And x_3 **is the positive root of the cubic equation**

$$f(x_3) = B^*x_3^3 - (B^*K + (C^*-B^*)j)x_3^2 + j((C^*-B^*)K + aB^*)x_3 + jK(D_2 - B^*a) = 0.$$

where

$$B^* = \frac{a_1}{\omega}\left(\frac{\omega_3}{c} + \frac{a_2\omega_3 b}{\omega_2 c} - 1\right), \quad C^* = \frac{a_1\omega_1\omega_3 b}{\omega\omega_2 c}.$$

The general Jacobian matrix is

$$J = \begin{bmatrix} a_1 - \frac{2a_1X}{K} - \frac{wY(2X/j+1)}{(\frac{X^2}{j}+X+a)^2} & -\frac{wX}{\frac{X^2}{j}+X+a} & 0 \\ Y\left[\frac{w_1(a-X^2/j)}{(\frac{X^2}{j}+X+a)^2}\right] & -a_2 + \frac{w_1X}{\frac{X^2}{j}+X+a} - \frac{w_2Z(D_2+bZ)}{(Y+D_2+bZ)^2} & Y\left[\frac{-w_2(Y+D_2)}{(Y+D_2+bZ)^2}\right] \\ 0 & Z\left[\frac{w_3(D_2+bZ)}{(Y+D_2+bZ)^2}\right] & -c + \frac{w_3Y(Y+D_2)}{(Y+D_2+bZ)^2} \end{bmatrix}$$

13.2 Stability Analysis:

- $E_0(0,0,0)$: **The trivial equilibrium $E_0(0,0,0)$ exists. The eigenvalues of the variational matrix around E_0 are $0,0,c$. Therefore, E_0 is a non hyperbolic point.**

- $E_1(K,0,0)$:**The equilibrium point $E_1(K,0,0)$ exists. The eigenvalues of the variational matrix around E_1 are $0,\pm c,\pm a_1$. Therefore, E_1 is also a nonhyperbolic point.**

- $E_2(x_2,y_2,0)$: **One eigenvalue is $-c+\frac{w_3y_3}{D_2+y_3}$, and the remaining two are given as**

$$-\frac{a_1x_3}{K}+\frac{jmx_3(j+2x_3)y_3}{2(aj+x_3(j+x_3))^2}\pm\sqrt{M}$$

where

$$\begin{aligned}M=x_3\left(aj+x_3(j+x_3)\right)^2\Bigg[&a_1^2x_3\left(aj+x_3(j+x_3)\right)^4\\&-2jwK\left(aj+x_3(j+x_3)\right)\Big\{aj\left(2jw_1K+a_1x_3j+2a_1x_3^2\right)\\&+x_3^2\left(-2jw_1K+a_1(j+x_3)(j+2x_3)\right)\Big\}y_3\\&-K^2w^2x_3(j+2x_3)^2y_3^2\Bigg].\end{aligned}$$

Hence, E_2 is asymptotically stable if $c>\frac{w_3y_3}{D_2+y_3}$ and

$$\frac{a_1x_3}{K}+\sqrt{M}>\frac{jwx_3y_3(j+2x_3)}{2(aj+x_3(j+x_3))^2},\quad\textbf{where } M>0.$$

- $E_3(x_3,y_3,z_3)$:

  The equilibrium point non-zero $E_3(x_3,y_3,z_3)$ exists under the following conditions: (i) $x_4<K$, (ii) $y_4>\frac{cD_2}{\omega_3-c}$, (iii)$\omega_3>c$, (iv)

$$D_2+\frac{aa_1}{\omega}<\frac{aa_1\omega_3}{\omega c}\left(1+\frac{a_2b}{\omega_2}\right).$$

  For $E_3(x_3,y_3,z_3)$, the characteristic equation of system (2.1) is given by

$$\lambda^3+\alpha\lambda^2+\beta\lambda+\gamma=0,\quad(2.3)$$

  where

$$\alpha=-(D+G+A),\quad\beta=DG-EF-BC+AD+AG,$$
$$\gamma=A(EF-DG)+BCG$$

and

$$A = \frac{\omega x_4 y_4 \left(\frac{2x_4}{j} + 1\right)^2}{\left(\frac{x_4^2}{j} + x_4 + a\right)} - \frac{a_1 x_4}{K},$$

$$B = -\frac{w x_4}{\left(\frac{x_4^2}{j} + x_4 + a\right)},$$

$$C = \frac{a w_1 y_4 - w_1 y_4 \frac{x_4^2}{j}}{2},$$

$$D = \frac{w_2 y_4 z_4}{(y_4 + D_2 + b z_4)^2},$$

$$E = -\frac{w_2 y_4^2 + D_2 w_2 y_4}{(y_4 + D_2 + b z_4)^2},$$

$$F = \frac{D_2 w_3 z_4 + b w_3 z_4^2}{(y_4 + D_2 + b z_4)^2},$$

$$G = -c + \frac{w_3 y_4^2 + D_2 w_3 y_4}{(y_4 + D_2 + b z_4)^2}.$$

Hence, by Routh–Hurwitz criterion, $E_3(x_3, y_3, z_3)$ is asymptotically stable when $\alpha, \beta, \gamma > 0$ and $\alpha\beta > \gamma$.

13.3 Numerical Simulation Results

We perform numerical computations to observe various dynamics of the co-existence equilibrium for the system (4.1). We consider the fixed parameter values as:

$$a_1 = 1, \quad k = 30, \quad w = 0.75, \quad j = 12, \quad a = 60, \quad a_2 = 0.05,$$

$$w_1 = 0.6, \quad w_2 = 0.2, \quad D_2 = 50, \quad b = 0.21, \quad c = 0.06, \quad w_3 = 0.12.$$

With these parameter values, we consider the initial value as $(6, 5, 3)$ for each simulation. In this case, the system (4.1) is asymptotically stable.

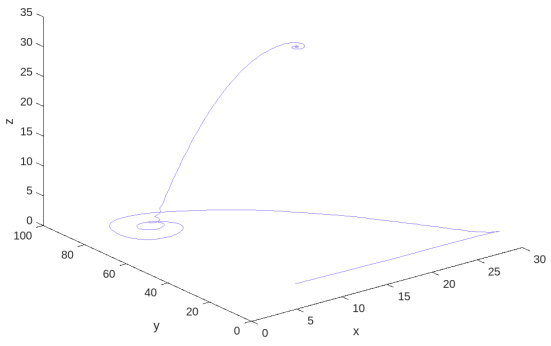


Figure 21: Timeseries of system

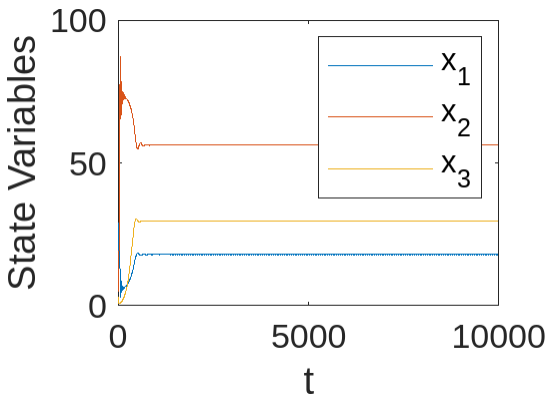


Figure 22: Phase-space diagram of system

13.4 Sensitivity to Initial Conditions

The trajectories corresponding to the two initial conditions (represented in blue and red) start close to each other but diverge over time. This suggests a high sensitivity to initial conditions, a hallmark of chaotic dynamics. Both trajectories exhibit frequent and abrupt drops in X, indicating potential transitions between different dynamical states. The occurrences of these drops appear irregular, further reinforcing chaotic behavior.Despite short-term divergence, both trajectories remain within a bounded range, suggesting that the system does not escape to infinity.

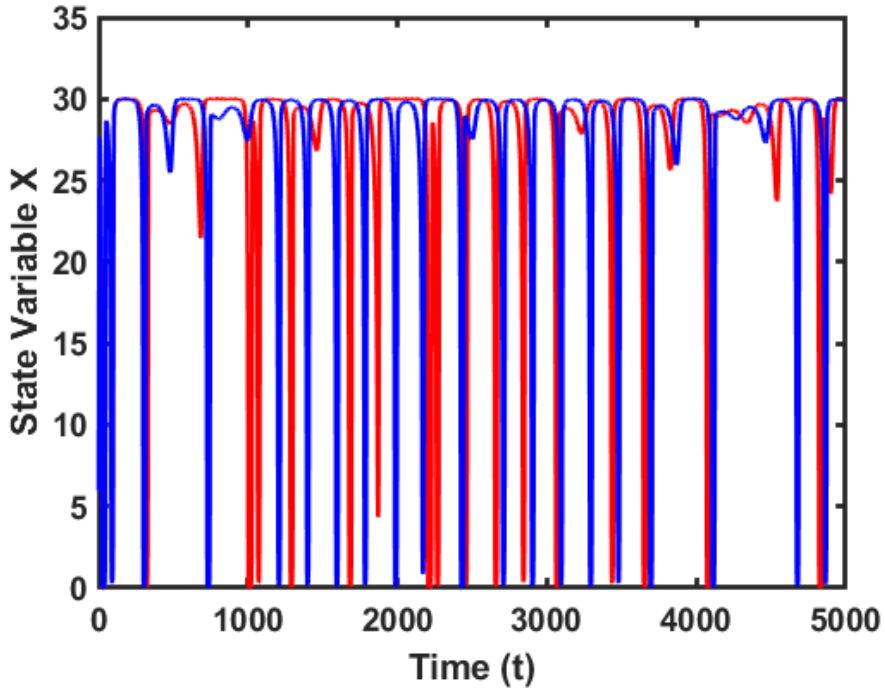


Figure 23: Time Series of x with respect to two different initial condition (6,5,3) and (6.21,5.13,3.03)

13.5 Chaotic Behavior

We have analytically studied the behavior of the system (4.1) when the values of some parameters were slightly changed. If we kept the values of the parameters $K, w, j, a, a_2, W_1, w_2, D_2, b, c$ unchanged, but change the value of the remaining parameter a_1 and w_3 plot the phase space diagram, we can see a chaotic behavior of the system.

Case-I If we keep the values of the parameters $K = 30, w = 0.75, j = 12, a = 20, a_2 = 0.08, w_1 = 0.89, w_2 = 0.5, D_2 = 50, b = 0.21, w_3 = 0.12, c = 0.01$ unchanged, but change the value of the remaining parameter a_1 and plot the phase space diagram with initial condition (6,5,3).

As a_1 increases, the system becomes more sensitive to initial conditions, especially in the lower range. However, beyond a certain point (around $a_1 0.5$,

the rate of increase in chaos saturates, suggesting a transition to a relatively stable chaotic regime.

The bifurcation diagram clearly indicates the route to chaos via period-doubling bifurcations as the parameter a_1 increases. The dense, scattered, and irregular distribution of x maxima for higher a_1 values, along with windows of periodicity amidst chaotic regions, confirms the existence of chaotic dynamics in the system.

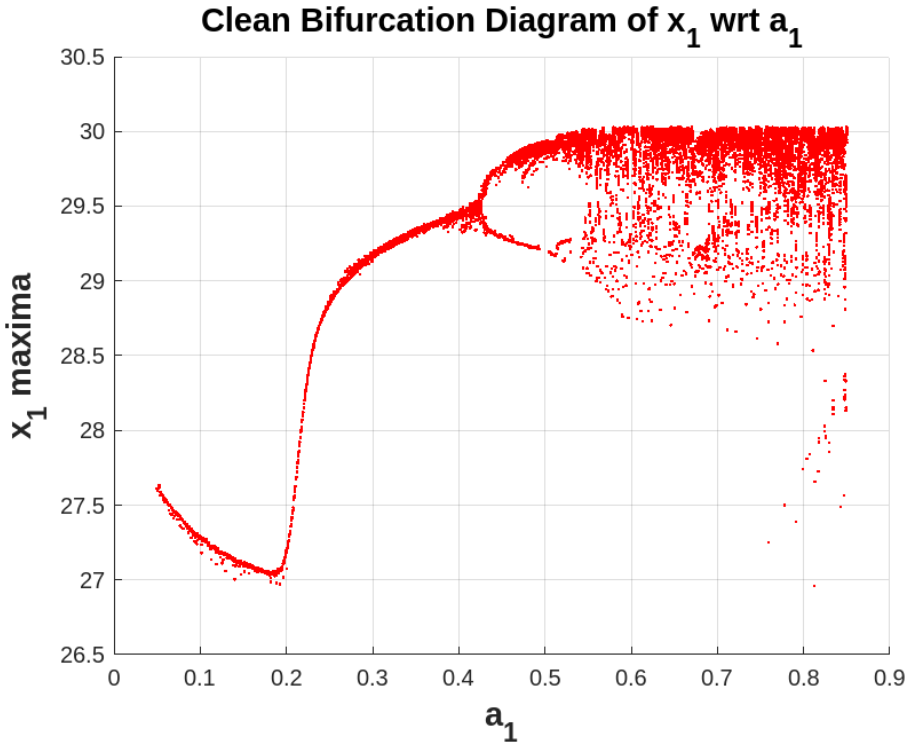


Figure 24: Bifurcation Diagram of x with respect to a_1

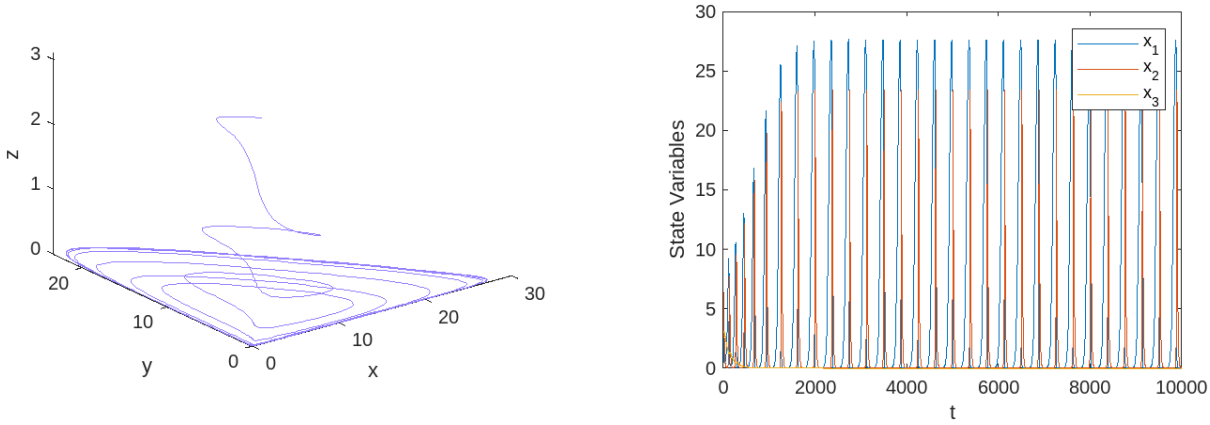


Figure 25: Timeseries (Right) and phase-space(Left) diagram of system for a_1=0.05

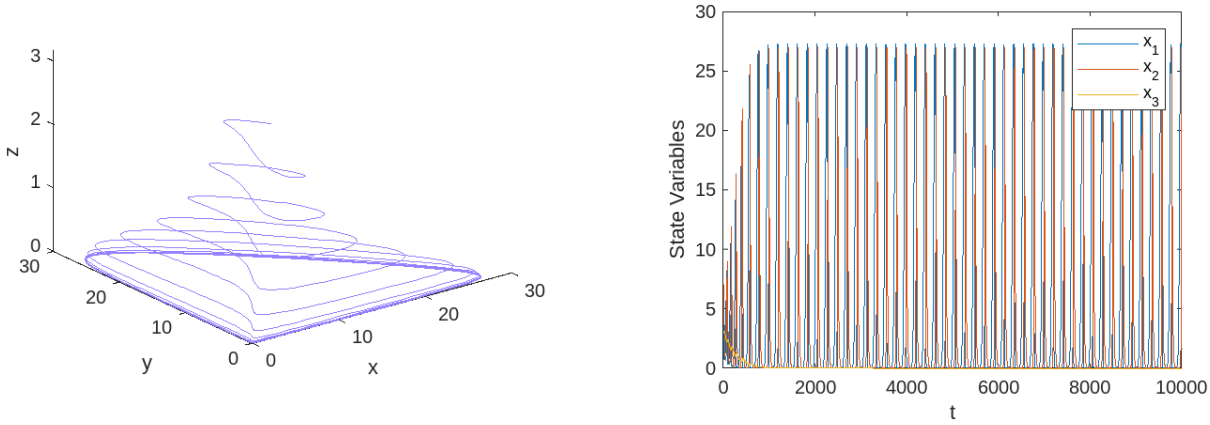


Figure 26: Timeseries (Right) and phase-space(Left) diagram of system for a_1=0.1

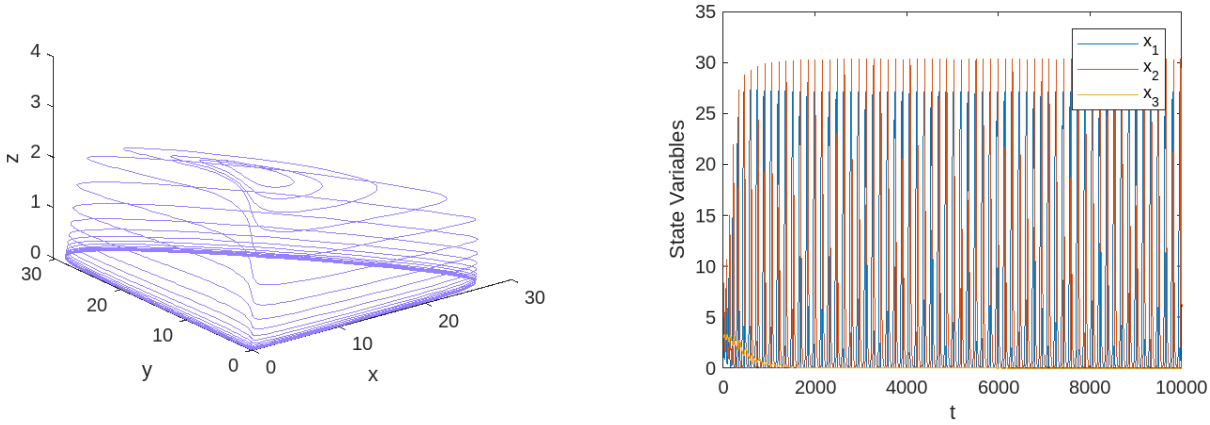


Figure 27: Timeseries (Right) and phase-space(Left) diagram of system for a_1=0.15

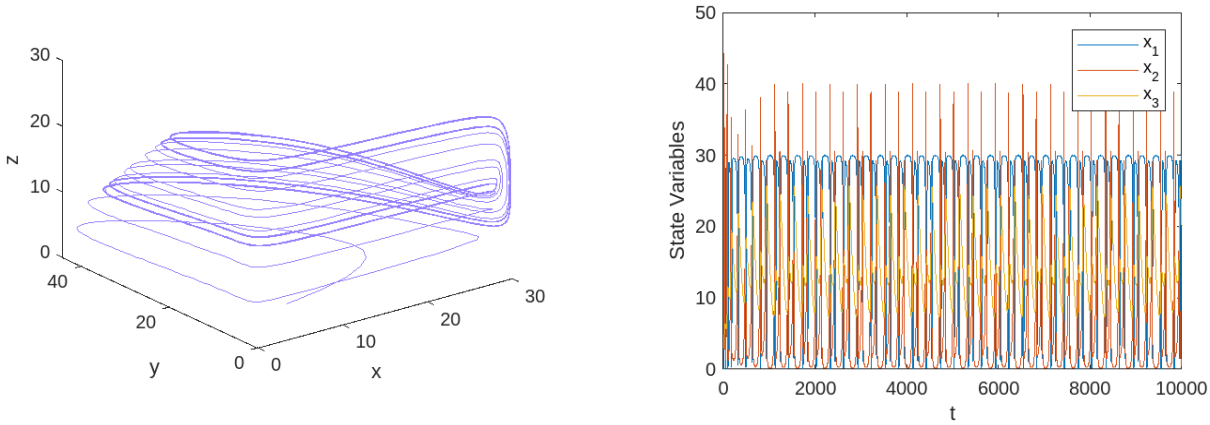


Figure 28: Timeseries (Right) and phase-space(Left) diagram of system for a_1=0.51

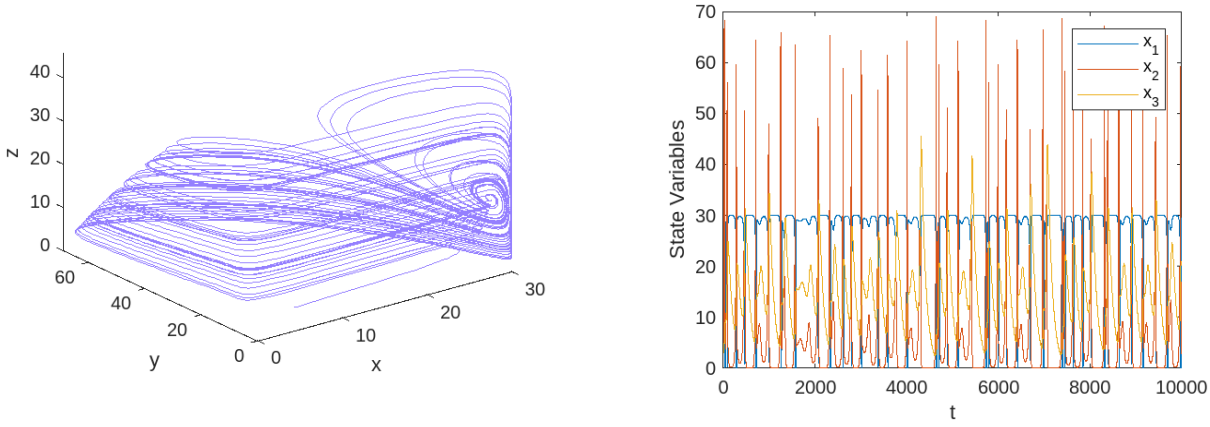


Figure 29: Timeseries (Right) and phase-space(Left) diagram of system for a_1=0.95

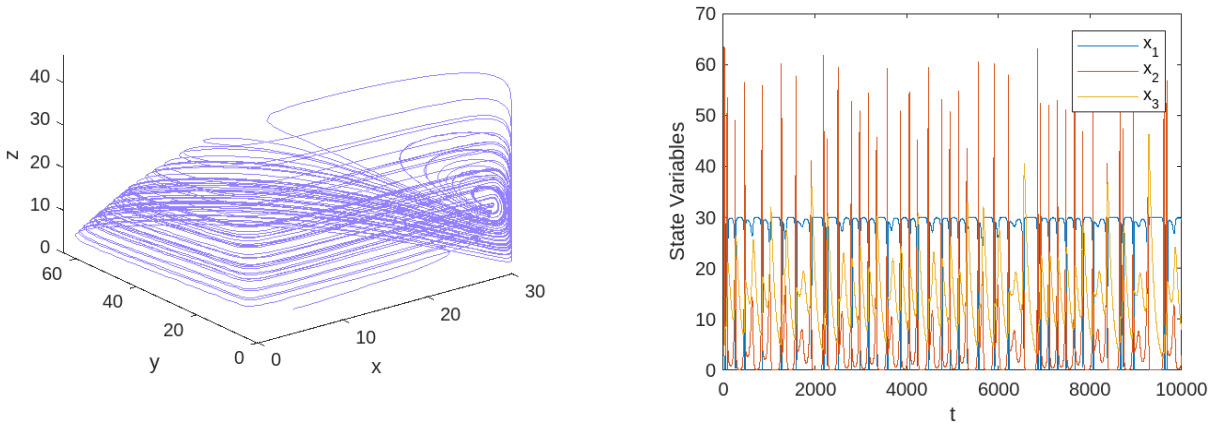


Figure 30: Timeseries (Right) and phase-space(Left) diagram of system for a_1=0.85

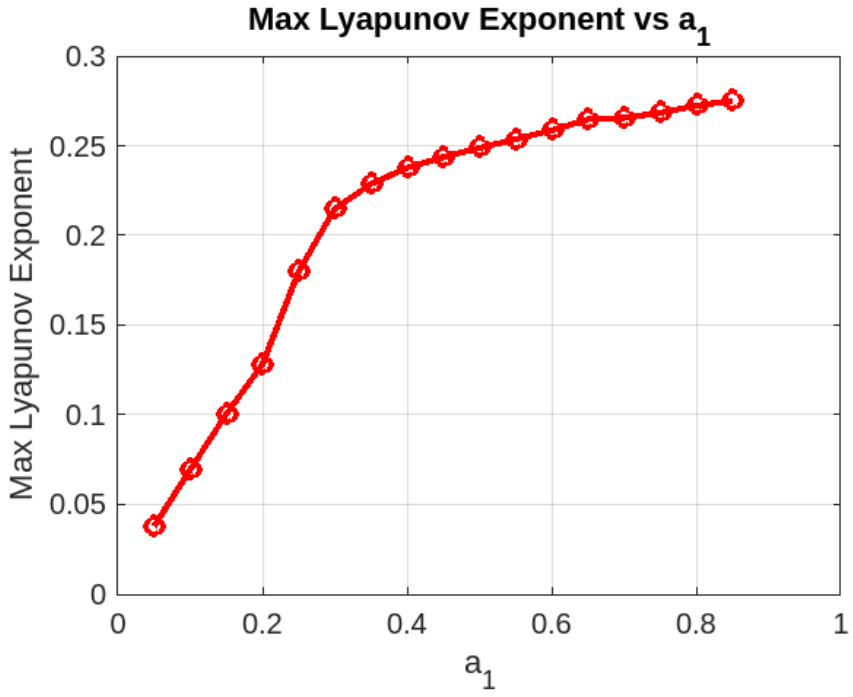


Figure 31: Lyapunov Exponent analysis for the parameter a_1

Case-II

If we kept the values of the parameters $a_1 = 1, K = 30, w = 0.75, j = 12, a = 20, a_2 = 0.08, w_1 = 0.89, w_2 = 0.5, D_2 = 50, b = 0.20, c = 0.12$ unchanged, but change the value of the remaining parameter w_3 and plot the phase space diagram with initial condition (6,5,3).

The parameter w_3 which likely influences predator-prey or top-level interaction strength, introduces complex nonlinear feedback that results in chaotic windows interleaved with more regular behavior. This suggests the system has a rich dynamical structure, with multiple transitions in the intensity of chaos as w_3 changes.

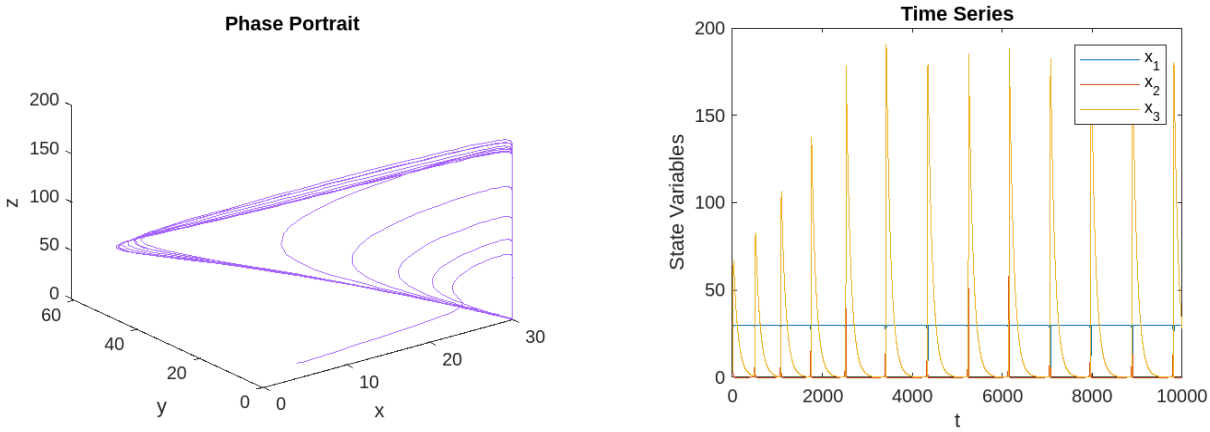


Figure 32: Timeseries (Right) and phase-space(Left) diagram of system for w_3=1.2

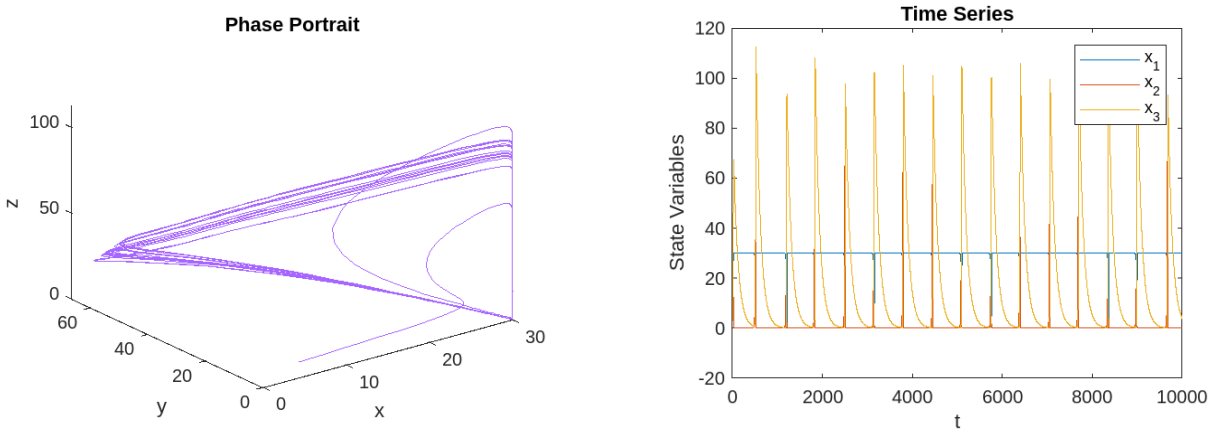


Figure 33: Timeseries (Right) and phase-space(Left) diagram of system for w_3=0.75

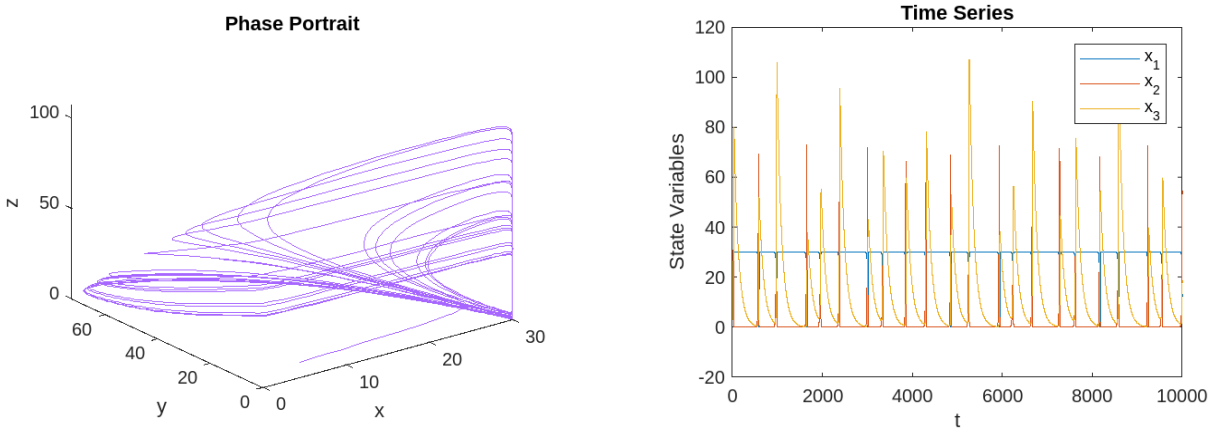


Figure 34: Timeseries (Right) and phase-space(Left) diagram of system for w_3=0.5

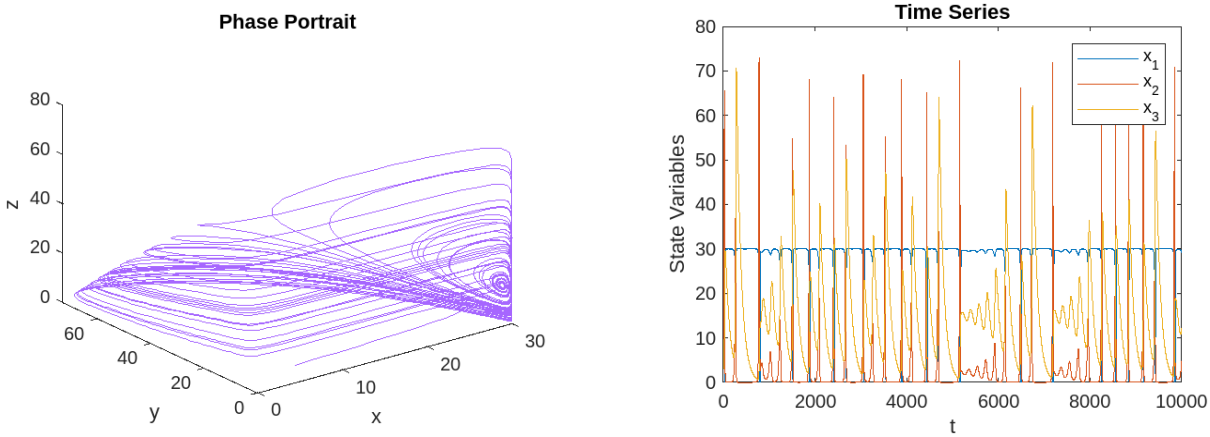


Figure 35: Timeseries (Right) and phase-space(Left) diagram of system for w_3=0.25

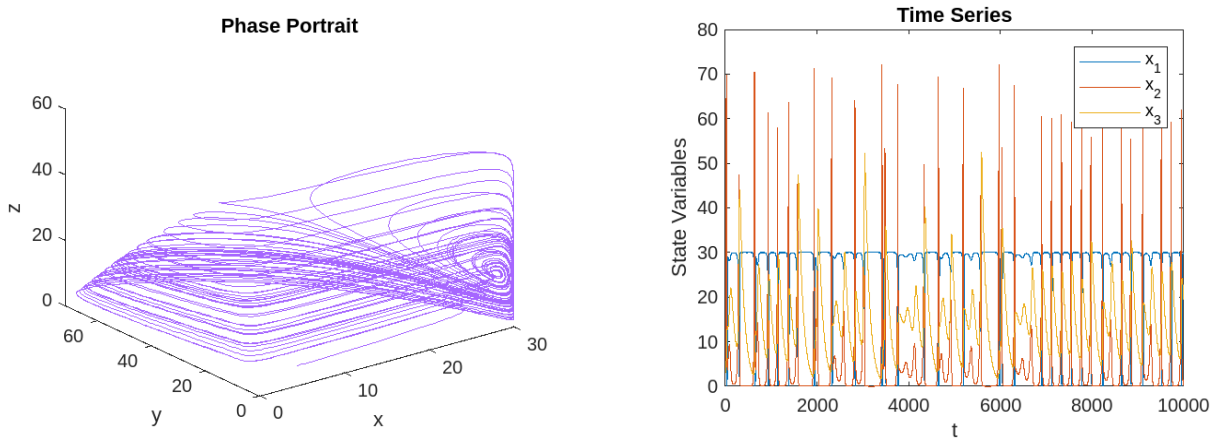


Figure 36: Timeseries (Right) and phase-space(Left) diagram of system for w_3=0.15

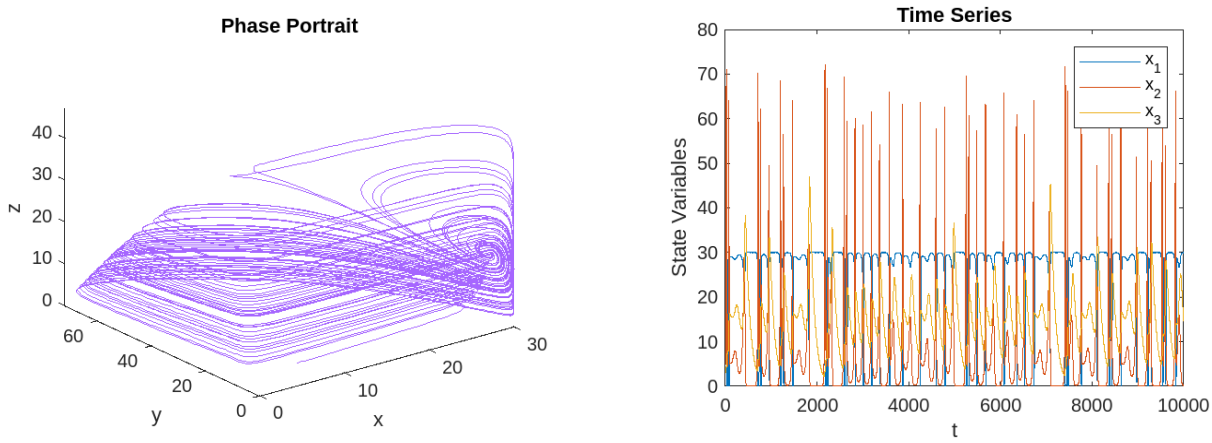


Figure 37: Timeseries (Right) and phase-space(Left) diagram of system for w_3=0.1

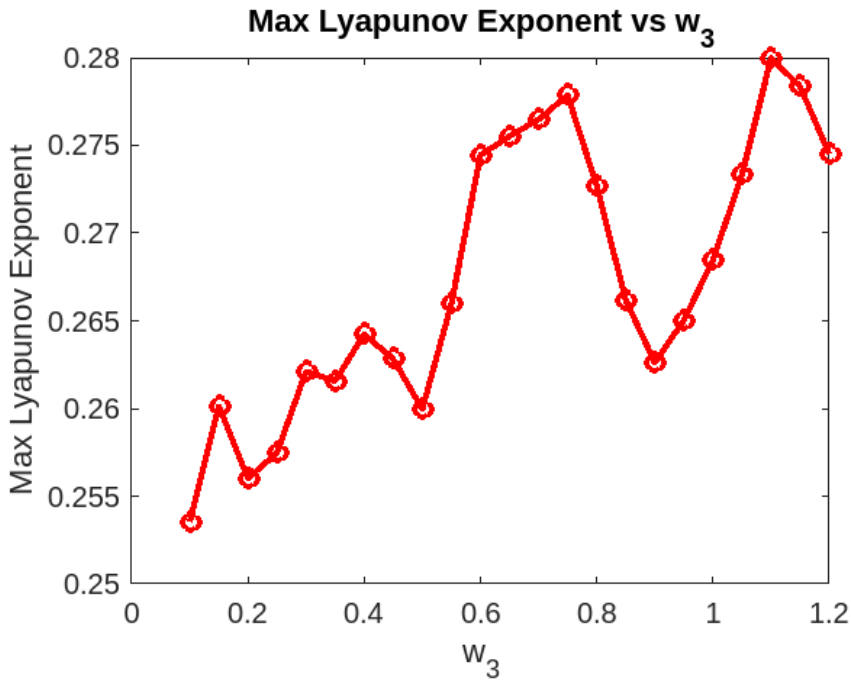


Figure 38: Lyapunov Exponent analysis for the parameter w_3

13.6 Applications & Implications of the Model

Predator–Prey–Top Predator System Mapping:

- $x(t)$: **Prey (e.g., Grass)**
- $y(t)$: **Predator (e.g., Deer)**
- $z(t)$: **Top predator (e.g., Tiger)**

Applications:

- **Maintaining biodiversity and preventing degradation of forest ecosystems.**
- **Managing human-wildlife conflict.**
- **Predicting ecological outcomes of predator removal or overhunting**
  - **Deer overpopulation**
  - **Vegetation depletion**
- **Simulating long-term impacts of ecological policies before actual implementation.**

Real life incident: In Yellowstone National Park, deer and elk overpopulation led to overgrazing, damaging vegetation and disrupting the ecosystem. Using chaos theory, ecologists modeled predator-prey-plant interactions, revealing that small changes could cause unpredictable, large-scale effects. A key example was the 1995 reintroduction of wolves, which reduced elk numbers and allowed vegetation to recover, benefiting other species like beavers and birds. This demonstrated sensitive dependence on initial conditions, a core idea in chaos theory. The incident showed how ecological systems can shift dramatically due to minor changes, making chaos theory valuable for understanding and managing complex natural environments like Yellowstone.

14 Conclusion

- **Nonlinear dynamical systems often display intricate behaviors, including stable, periodic, and chaotic dynamics.**
- **Chaos, though deterministic, is highly sensitive to initial conditions, making long-term predictions challenging.**
- **The studied three-species model reveals how nonlinear interactions can give rise to rich and unpredictable dynamics, verified through equilibrium analysis, bifurcation, and phase portraits.**
- **Here in the three species model we see that the model is chaotic in the range of a_1 from 0.05 to 0.85 and for w_3 from 0.1 to 1.2 with the other parameters explained before.**

15 References

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- **Debaldev Jana , Rashmi Agrawal , Ranjit Kumar Upadhyay, 2014 ; Top-predator interference and gestation delay as determinants of the dynamics of a realistic model food chain**
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- **Hena Rani Biswas et al., CHAOS THEORY AND ITS APPLICATIONS IN OUR REAL LIFE, Barishal University Journal Part 1, 5(1 and 2): 123-140 (2018)**